

**BANARAS HINDU UNIVERSITY**

B.A./B.Sc. (Hons.) Semester V Examination, 2016-17

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**Paper No. STA/STB-504: Operations Research**

Time: 3 Hours

Maximum Marks: 70

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*Instructions: Answer any **five** questions. All questions carry equal marks. Symbols have their usual meanings.*

**Question 1.**

Explain how and why Operations Research (OR) methods have been valuable in aiding executive decision-making. [14]

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**Question 2.**

(a) Define the following terms:

(i) Non-degenerate and degenerate solution

(ii) Optimum solution

[7]

(b) Prove that the objective function of an LPP assumes its minimum value at an extreme point of the convex set of feasible solutions. If it assumes its minimum at more than one extreme point, then it takes on the same value for every convex combination of those points. [7]

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**Question 3.**

Give a brief account of the simplex method for solving an LPP. Solve the following LPP by the simplex method:

$$\begin{array}{ll}\text{Maximise} & Z = 3x_1 - x_2 \\ \text{subject to} & x_1 + x_2 \leq 4 \\ & x_1 + 3x_2 \leq 6 \\ & x_2 \leq 4 \\ & x_1, x_2 \geq 0\end{array}$$

[14]

**Question 4.**

Show that the transportation problem can be regarded as a particular case of an LP problem. Solve the following transportation problem and find the optimum solution:

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|             | $D_1$ | $D_2$ | $D_3$ | $D_4$ | Availability |
|-------------|-------|-------|-------|-------|--------------|
| $O_A$       | 1     | 7     | 2     | 5     | 5            |
| $O_B$       | 1     | 4     | 6     | 5     | 15           |
| $O_C$       | 6     | 3     | 3     | 2     | 10           |
| $O_D$       | -1    | 4     | 2     | –     | 20           |
| Requirement | 20    | 25    | 15    | –     |              |

[14]

**Question 5.**

What do you mean by inventory problems? Describe the ELS (Economic Lot Size) model with a uniform rate of demand, finite rate of production, and no shortages. An item is produced at the rate of 50 items per day. The demand occurs at the rate of 25 items per day. If the setup cost is Rs. 100 per setup and the holding cost is Rs. 0.01 per unit per day, find the ELS for one run, assuming shortages are not permitted. [14]

**Question 6.**

What do you understand by duality in LPP? What are the standard forms of the dual? Write the dual of the following LPP and solve it. Also find the primal solution:

$$\begin{aligned}
 &\text{Minimise } Z = 0.07x_1 + 0.05x_2 \\
 &\text{subject to } 0.1x_1 \geq 0.4 \\
 &\quad x_1 \geq 0.6 \\
 &\quad x_1 + 0.2x_2 \geq 2.0 \\
 &\quad 2x_1 + 0.1x_2 \geq 1.8 \\
 &\quad x_1, x_2 \geq 0
 \end{aligned}$$

[14]

**Question 7.**

- (a) Explain saddle point and zero-sum game. For a  $2 \times 2$  zero-sum two-person game where the optimal strategies are mixed, show that the optimal strategies  $(x_1^*, x_2^*)$  for Player A and  $(y_1^*, y_2^*)$  for Player B are:

$$x_1^* = \frac{a_{22} - a_{21}}{(a_{11} + a_{22}) - (a_{12} + a_{21})}, \quad y_1^* = \frac{a_{22} - a_{12}}{(a_{11} + a_{22}) - (a_{12} + a_{21})},$$

and the value of the game to A is:

$$V = \frac{a_{11}a_{22} - a_{12}a_{21}}{(a_{11} + a_{22}) - (a_{12} + a_{21})}.$$

[7]

- (b) Solve the following game by the dominance property:

| Player A \ Player B | $B_1$ | $B_2$ | $B_3$ | $B_4$ |
|---------------------|-------|-------|-------|-------|
| $A_1$               | 2     | 3     | 4     | 5     |
| $A_2$               | 1     | 3     | 2     | 7     |
| $A_3$               | 3     | 4     | 8     | 6     |
| $A_4$               | 6     | 5     | 7     | 6     |
| $A_5$               | 2     | 0     | 8     | 1     |

[7]

**Question 8.**

Solve the following assignment problem to minimise total time (rows = jobs, columns = typists):

|       | $T_1$ | $T_2$ | $T_3$ | $T_4$ |
|-------|-------|-------|-------|-------|
| $J_1$ | 2     | 10    | 9     | 7     |
| $J_2$ | 15    | 4     | 14    | 8     |
| $J_3$ | 13    | 14    | 16    | 11    |
| $J_4$ | 4     | 15    | 13    | 9     |

[14]